

BeautifulSoup Documentation Analytics System

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1 Prerequisites

```
[1]: # This cell is for importing needed modules
from pathlib import Path
import sys

project_root = Path("..").resolve() # one level up from notebooks
sys.path.append(str(project_root))

import pandas as pd

from src import main, analyzer, visualizer
```

2 Dataset Overview

The dataset used in this project consists of the official BeautifulSoup documentation webpage. The HTML page was downloaded directly from the BeautifulSoup documentation website and stored locally for analysis. The purpose of the dataset is to examine the structure and content of the documentation by extracting textual information, hyperlinks, and Python code examples.

The extracted dataset contains three main components:

- Sections: Documentation headings, text content, hierarchy, word counts
- Links: Internal and external hyperlinks found in the documentation
- Code Examples: Python code snippets contained in the documentation

The processing pipeline successfully extracted 111 documentation sections.

3 Scraping Method

The project follows a complete web scraping pipeline implemented using Python.

3.1 Download HTML

The documentation page was downloaded using an HTTP request and saved as a local HTML file.

```
[2]: main.download_html()
```

FEATURE 1 - WEB PAGE COLLECTOR

HTTP Status Code: 200

Download completed.

Output:

- HTTP Status Code: 200
- HTML downloaded successfully

3.2 Parse HTML

The downloaded HTML document was parsed using the BeautifulSoup library to create a navigable document object.

```
[3]: main.parse_html(verbose=True); # Semicolon to surpress output
```

FEATURE 2 - HTML PARSER

HTML parsed successfully.

3.3 Extract Data

Three extraction modules were implemented:

- Section extractor
- Link extractor
- Code example extractor

Each extracted dataset was exported into a CSV file for later analysis.

3.3.1 Sections

```
[4]: main.sections_to_csv()
```

3.3.2 Links

```
[5]: main.links_to_csv()
```

3.3.3 Code Examples

```
[6]: main.code_samples_to_csv()
```

4 Extracted Data Summary

The scraping process produced structured datasets describing the documentation.

Documentation Statistics

- Number of Sections: 111
- Internal Links: 396

- External Links: 30
- Average Section Length: 363.04 words
- Longest Section: Searching the tree
- Maximum Word Count: 4612 words

The largest section is Searching the tree, containing:

- 4612 words
- 189 code examples
- 93 hyperlinks

making it the most comprehensive section in the documentation.

5 Analysis Results

```
[7]: # This cell to load data for analysis
analyzer.load_data()
```

5.1 Number of Sections

```
[8]: print(f"Number of sections: {int(analyzer.sections_count())}")
```

Number of sections: 111

The documentation contains 111 sections, indicating a large and well-organized knowledge base.

5.2 Section with the highest word count

```
[9]: df = pd.DataFrame(analyzer.section_with_longest_count())
df
```

```
[9]:
```

section_id	searching-the-tree	29
section_level		1
section_title	Searching the tree	
section_text	Beautiful Soup defines a lot of methods for se...	
word_count		4612
code_block_count		189
link_count		93

5.3 Section with the most code examples

```
[10]: df = pd.DataFrame(analyzer.section_with_most_code_examples())
df
```

```
[10]:
```

section_id	searching-the-tree	29
section_level		1
section_title	Searching the tree	
section_text	Beautiful Soup defines a lot of methods for se...	
word_count		4612
code_block_count		189
link_count		93

5.4 Section with the most links

```
[11]: df = pd.DataFrame(analyzer.section_contains_most_links())
df
```

```
[11]:
```

section_id	searching-the-tree	29
section_level		1
section_title	Searching the tree	
section_text	Beautiful Soup defines a lot of methods for se...	
word_count		4612
code_block_count		189
link_count		93

The section Searching the tree is the longest section in the documentation.

Statistics:

- Word count: 4612
- Code examples: 189
- Links: 93

This suggests that searching operations are the core functionality of BeautifulSoup and therefore require more detailed explanations and examples.

5.5 Most frequent technical keywords

The ten most frequently occurring technical keywords are:

```
[12]: top_ten = analyzer.top_ten_technical_keyword()
df = pd.DataFrame(top_ten)
df.columns= ["Keyword", "Count"]
df.index = range(1, len(df) + 1) # So index starts from 1
df
```

```
[12]:
```

	Keyword	Count
1	soup	1047
2	tag	928
3	class	559
4	html	505

5	string	425
6	example	419
7	link	413
8	find	395
9	com	383
10	href	378

These keywords indicate that the documentation primarily focuses on HTML parsing, tags, searching, and hyperlinks.

5.6 Link Analysis

```
[13]: internal_links, external_links = analyzer.count_internal_and_external_links()
print(f"{internal_links} internal links and {external_links} external links.")
```

396 internal links and 30 external links.

This indicates that the documentation is highly interconnected, allowing readers to navigate between related topics while relying relatively little on external resources.

5.7 Code Example Analysis

```
[14]: print(f"{analyzer.count_find_all()} examples use find_all()")
```

39 examples use find_all()

```
[15]: print(f"{analyzer.count_get_text()} examples use get_text()")
```

4 examples use get_text()

This demonstrates that find_all() is one of the most frequently demonstrated BeautifulSoup methods.

5.8 Section Length Statistics

Section lengths vary considerably.

```
[16]: average, minimum, maximum, std_dev = analyzer.word_count_statistics()
print(f"Average: {average:.2f}")
print("Minimum:", minimum)
print("Maximum:", maximum)
print(f"Standard deviation: {std_dev:.2f}")
```

Average: 363.04

Minimum: 24

Maximum: 4612

Standard deviation: 557.83

The large standard deviation indicates substantial variation in section sizes, with some sections providing brief explanations while others serve as extensive tutorials.

5.9 Longer than average sections

```
[17]: df2 = analyzer.sections_above_average()
df2.columns = ["Title", "Word Count"]
df2
```

```
[17]:
```

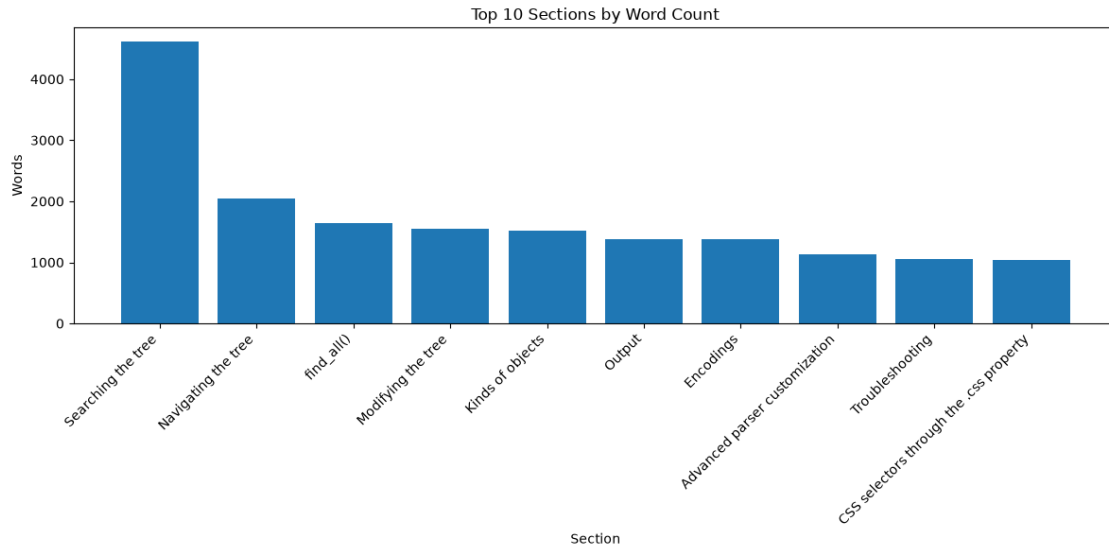
	Title	Word Count
3	Quick Start	422
4	Installing Beautiful Soup	427
7	Kinds of objects	1515
8	Multi-valued attributes	414
12	Navigating the tree	2051
13	Going down	890
23	Going sideways	454
26	Going back and forth	378
29	Searching the tree	4612
30	Kinds of filters	597
36	find_all()	1646
38	The keyword arguments	500
39	Searching by CSS class	381
50	CSS selectors through the .css property	1039
54	Modifying the tree	1547
69	Output	1383
72	Output formatters	870
76	Specifying the parser to use	635
77	Differences between parsers	403
78	Encodings	1381
80	Unicode, Dammit	596
86	Low-level search interface	712
87	Custom element filtering	629
89	Advanced parser customization	1125
90	Parsing only part of a document	546
94	Troubleshooting	1048
103	Beautiful Soup 3	869
104	Porting code to BS4	759

6 Visualization Charts

6.1 Top 10 sections by word count

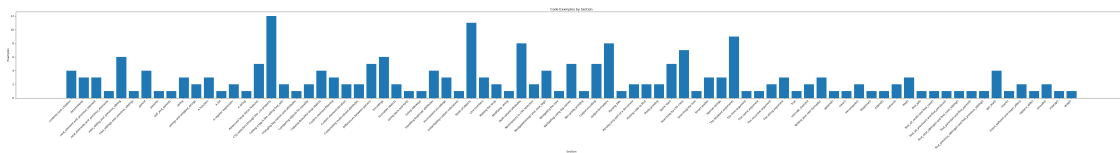
```
[18]: # This cell to load data for visualization
visualizer.initializer()
```

```
[19]: visualizer.chart_top10_sections()
```



6.2 Number of code examples by section

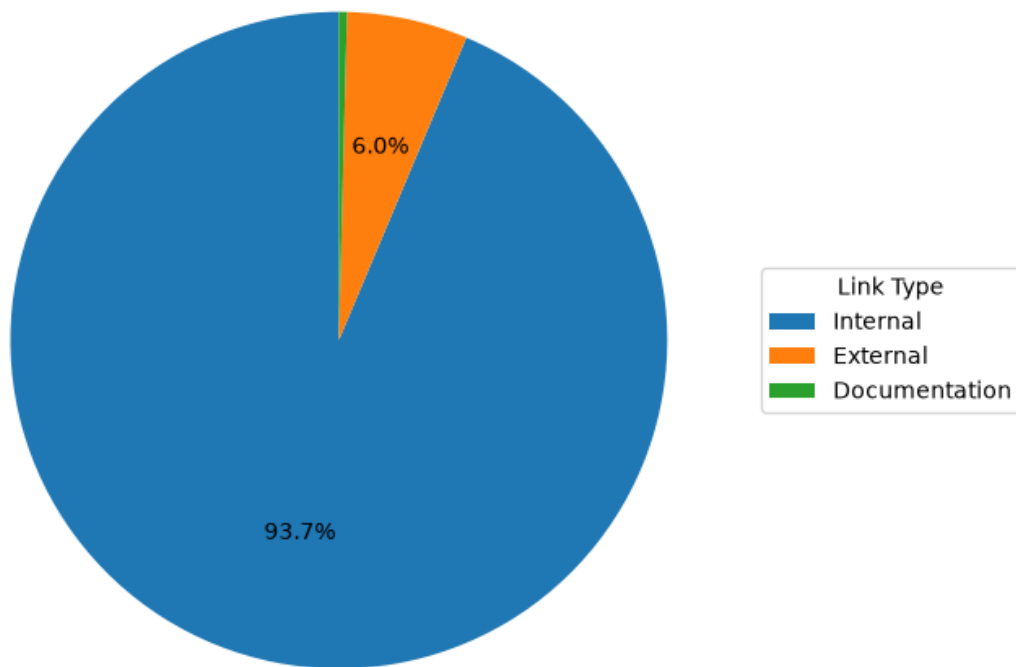
```
[20]: visualizer.chart_code_examples()
```



6.3 Link type distribution

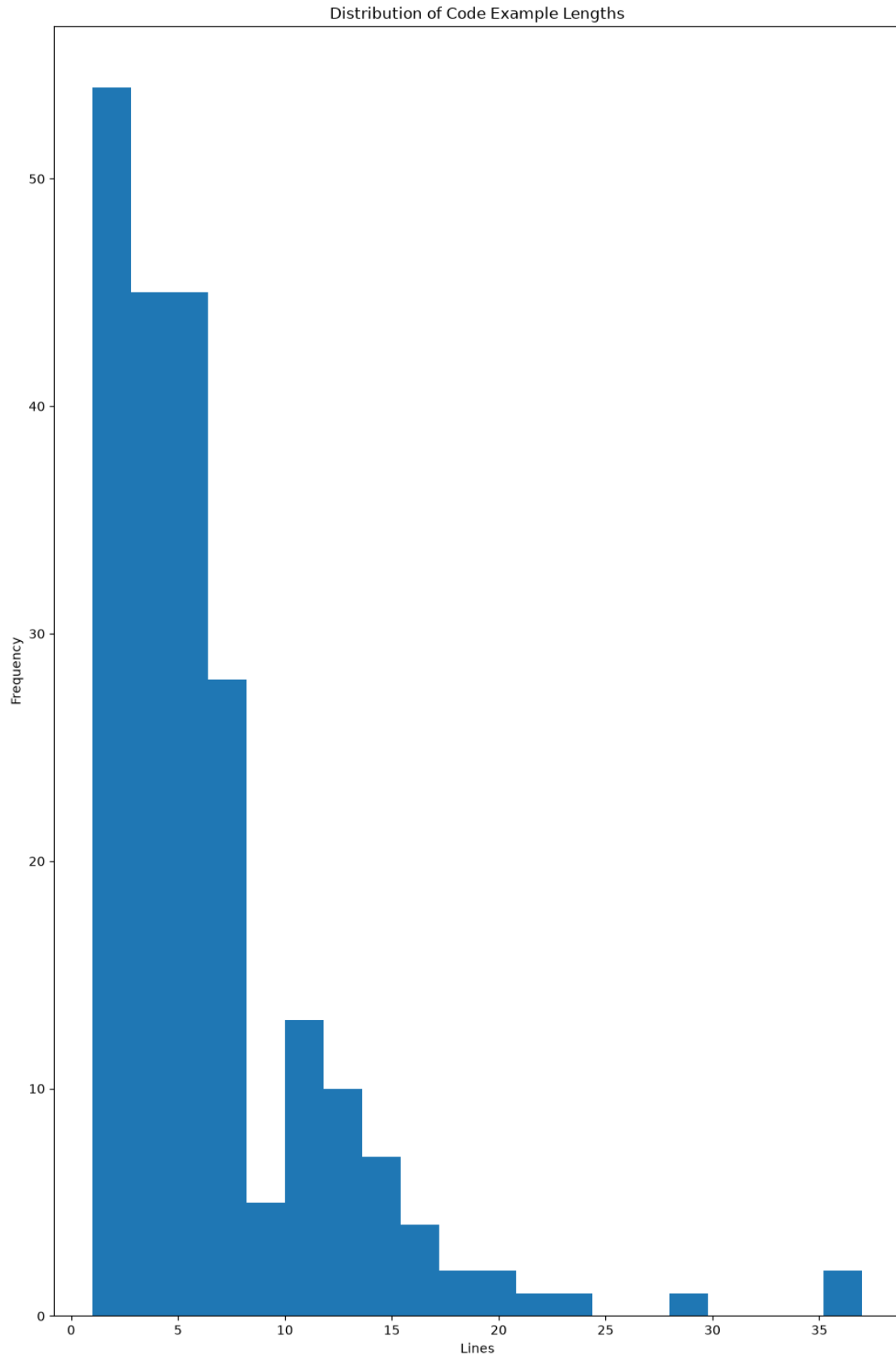
```
[21]: visualizer.chart_link_distribution()
```

Link Type Distribution



6.4 Code example line count distribution

```
[22]: visualizer.chart_code_length_histogram()
```

7 Key Findings

The project reveals several insights about the BeautifulSoup documentation:

- The documentation contains 111 structured sections covering a wide range of topics.
- Searching the tree is the largest and most comprehensive section, suggesting that searching functionality is central to BeautifulSoup.
- Internal hyperlinks significantly outnumber external hyperlinks, making the documentation highly interconnected.
- The keyword analysis confirms that the documentation emphasizes HTML parsing, tags, and searching operations.
- `find_all()` appears much more frequently than `get_text()`, reflecting its importance in common BeautifulSoup workflows.
- Documentation section lengths vary widely, with a few detailed tutorial sections and many shorter reference sections.

8 Limitations

Several limitations should be considered:

- The analysis is based on a single snapshot of the BeautifulSoup documentation; future updates may change the results.
- Only one webpage was analyzed, so the findings cannot be generalized to other documentation websites.
- Keyword frequency does not capture semantic meaning or context.
- Code examples were analyzed mainly by counting occurrences rather than evaluating their complexity or functionality.
- Dynamic content loaded through JavaScript would not be captured by this scraping approach (though this is not a major issue for the BeautifulSoup documentation).

9 Conclusion

This project successfully developed an end-to-end documentation analytics system using Python and BeautifulSoup. The system automatically downloaded, parsed, extracted, analyzed, and visualized information from the BeautifulSoup documentation.

The analysis showed that the documentation is well structured, heavily interconnected through internal links, and centered around searching-related functionality. Automated extraction and visualization provided valuable insights into documentation organization, keyword usage, code examples, and content distribution. Overall, the project demonstrates how web scraping and data analysis techniques can be used to better understand and evaluate technical documentation.